# PERFORMANCE EVALUATION OF A SECURED BLE MESH NETWORK

## 1. Development Environment Setup

### 1.1 Installed Required Tools

nRF Connect for Desktop

Official tool for Nordic development.

https://www.nordicsemi.com/Products/Development-tools/nRF-Connect-for-desktop

**Installed the following applications:**

Toolchain Manager

Programmer

Serial Terminal

Bluetooth Low Energy (BLE)

**J-Link Driver**

**Required for flashing and debugging Nordic boards.**

https://www.segger.com/downloads/jlink

**Visual Studio Code**

**Code editor for development.**

**https://code.visualstudio.com/**

**nRF Connect for VS Code Extension**

**Installed extension inside VS Code for:**

Build

Flash

Debug

SDK management

https://www.nordicsemi.com/Products/Development-tools/nRF-Connect-for-VS-Code

**Installed nRF Connect SDK (v2.9+)**

**Using Toolchain Manager, installed:**

nRF Connect SDK v2.9.2

Compatible toolchain

## 2. Initial Board Testing

### 2.1 Blinky Test

**Created first project: Blinky**

**Verified:**

Board power supply

GPIO functionality

Build & flash working

Toolchain configured properly

This confirmed hardware and environment were correctly set up**.**

## 3. BLE Communication Testing

### 3.1 Kit-to-Kit Communication

**Configured:**

One board as Peripheral

Second board as Central

**Verified:**

Successful BLE connection

Data transfer between two nRF5340 DK boards

### 3.2 Mobile-to-Board Communication

**Used nRF Connect mobile app:**

 Android: https://play.google.com/store/apps/details?id=no.nordicsemi.android.mcp 

iOS: https://apps.apple.com/app/nrf-connect/id1054362403

**Tested:**

BLE advertising

Connecting to peripheral

Sending and receiving data

Confirmed BLE stack working correctly.

## 4. Hardware Integration

### 4.1 I2C 16x2 LCD Integration (Central Side)

**Connected:**

16x2 LCD with I2C module

Used I2C (TWI) peripheral

**Verified LCD working using test program:**

(HELLO NITESH )

**This confirmed:**

I2C configuration correct

SDA/SCL mapping correct

Power supply stable

Display functioning properly

## 4.2 **4x4 Matrix Keypad Integration (Peripheral Side)**

**Connected:**

4 Row pins

4 Column pins

Configured GPIO scanning logic

**Implemented:**

Row scanning

Column detection

Debounce handling

**Verified:**

Correct key detection

Key value printing via UART

Keypad data sent over BLE

# 5. **BLE Data Transmission from Keypad to LCD**

System Design:

Peripheral (Keypad)

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BLE Transmission

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Central (LCD Display)

**Working:**

Key pressed on keypad

Sent via BLE

Displayed on LCD in real time

**This demonstrated:**

BLE NUS Service implementation

GATT communication

Real-time wireless data transfer

6.**Secure PIN Authentication Implementation**

**Extended project to create:**

 **Secure PIN-Based Access System Logic:**

User enters PIN via keypad

PIN sent over BLE

Central device verifies PIN

**LCD displays:**

**If incorrect:**

**WRONG PIN**

**If correct:**

**HELLO**

**7. Real-World Applications**

**This system can be applied in:**

Smart Door Lock Systems

ATM PIN Authentication

Digital Lockers

Access Control Systems

Smart Home Security

Secure IoT Devices

**8.Technical Concepts Used**

**Embedded Systems**

GPIO configuration

I2C communication

UART debugging

Interrupt handling

**BLE Stack**

Peripheral role

Central role

GATT services

Nordic UART Service (NUS)

Secure pairing

**RTOS Concepts (Zephyr)**

Threads

Work queue

Heap memory

Kconfig configuration

**Debugging**

Build errors

Devicetree issues

SDK compatibility fixes

Flash memory settings

## 9.**Project Architecture Summary**

**Peripheral Device:**

4x4 Keypad

BLE Peripheral

Sends PIN

**Central Device:**

BLE Central

16x2 LCD via I2C

PIN verification logic

Output display

10. **Skills Demonstrated**

BLE protocol implementation

Embedded C development

Hardware interfacing

Real-time debugging

SDK migration handling

System integration

Secure authentication logic design

11. **Project Difficulty Level**

Intermediate to Advanced Embedded IoT System

**Includes:**

Multi-device communication

Wireless protocol

Hardware integration

Security logic implementation

12. **Why This is a Strong Resume Project**

**Because it combines:**

BLE

Security

Embedded Hardware

Real-world application

Nordic SDK

Zephyr RTOS

This is significantly better than a simple LED or UART project.

Workflow of Secure BLE-Based PIN Authentication System